

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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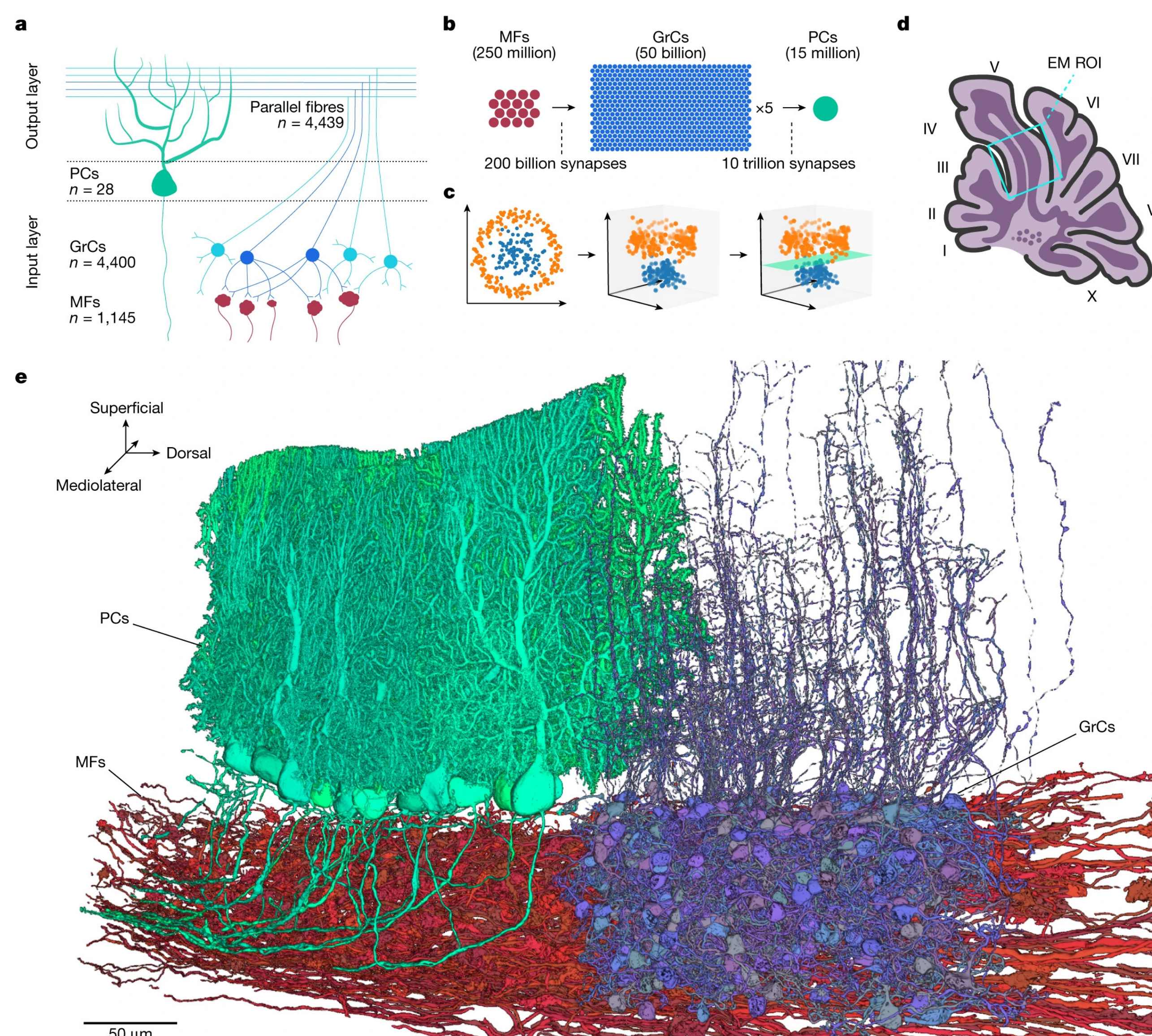
Introduction

1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC).
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to software incompatibilities and complex dependencies
3. We address this issue using singularity containers that encapsulate the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC and cloud systems.
4. We give a practical demonstration by benchmarking HPC environments with mossy fiber–granule cell network simulations.

Containerization for Scientific Computing

Container is a lightweight, standalone, and executable package
It is designed to include everything needed to run a scientific software

Benchmarking Data and Network*: Cerebellar Mossy Fiber → Granule Cell



* Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. Nature 613, 543–549 (2023)

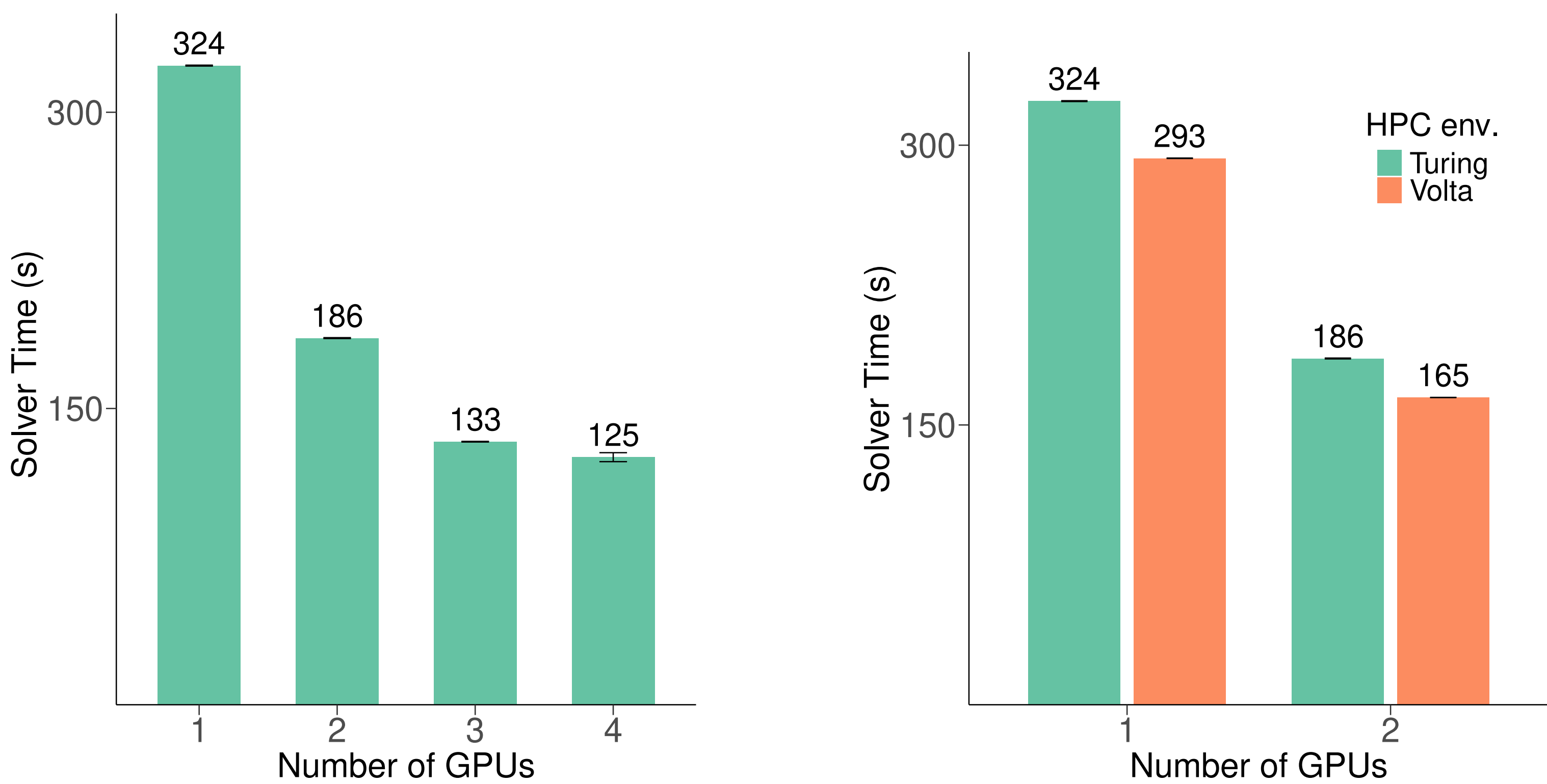
Table: Network Configuration and Simulation Parameters

Number of MF	85,600
Number of GRC	314,000
Number of compartments	942,000
Number of synapses	990,960
Simulation time	2 s

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU



(a) Turing architecture: Comparison of solver time for varying numbers of GPUs in CORENeuron-GPU simulator

(b) Comparison of solver time between HPC environments

Figure: Solver time comparisons in CORENeuron simulations across HPC platforms.

Turing, Volta outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.
Volta GPUs are faster than Turing at equivalent GPU counts, offering better efficiency per device.

Acknowledgment

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Our Computational Framework

